

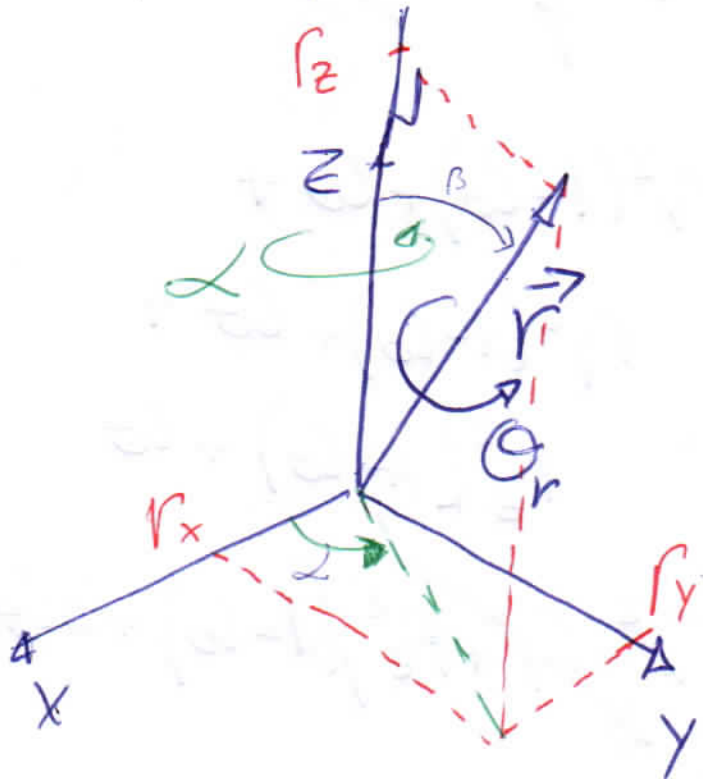
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①

$$= \text{Euler} \rightarrow z_1 y_2 z_3 \Rightarrow R_{z_1} R_{y_2} R_{z_3}$$

$$= \text{RPY} \rightarrow x \ y \ z \Rightarrow R_z R_y R_x$$

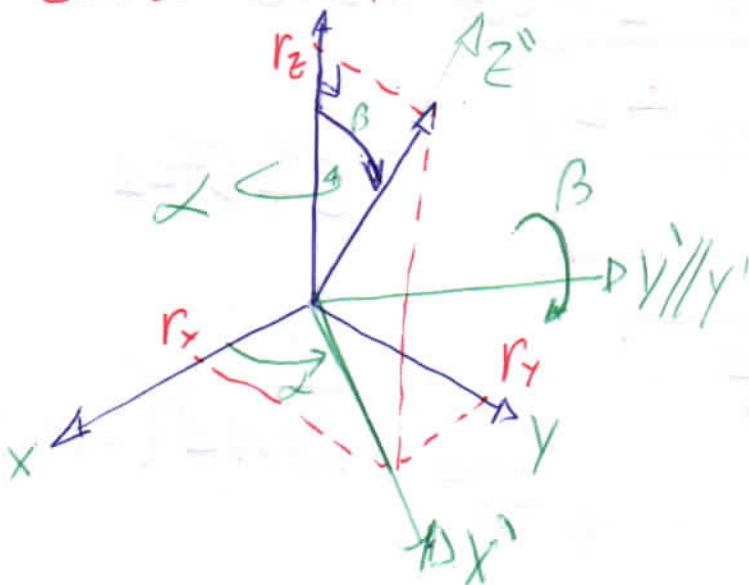
Rotac o em torno de um eixo



$$R_{\vec{r}}(\theta_r) = ?$$

$$\vec{r} = \begin{bmatrix} r_x \\ r_y \\ r_z \end{bmatrix} \quad \|\vec{r}\| = 1$$

"Levar z pl o eixo $\vec{r}$ "



$$R_z(\alpha) R_y(\beta)$$

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$$R_{\vec{r}}(\theta) = R_z(\alpha) R_y(\beta) R_z(\theta_r) R_y(-\beta) R_z(\alpha) \quad (2)$$

$$\alpha = ?$$

$$\beta = ?$$

$$C_\alpha = \frac{r_x}{\sqrt{r_x^2 + r_y^2}}$$

$$S_\alpha = \frac{r_y}{\sqrt{r_x^2 + r_y^2}}$$

$$C_\beta = r_z$$

$$S_\beta = \sqrt{r_x^2 + r_y^2}$$

$$R_{\vec{r}}(\theta) = \begin{bmatrix} r_x^2(1-\cos\theta) + C_\theta & r_x r_y(1-\cos\theta) - r_z S_\theta & r_x r_z(1-\cos\theta) - r_y S_\theta \\ r_x r_y(1+\cos\theta) + r_z S_\theta & r_y^2(1-\cos\theta) + C_\theta & r_y r_z(1-\cos\theta) + r_x S_\theta \\ r_x r_z(1-\cos\theta) - r_y S_\theta & r_y r_z(1-\cos\theta) + r_x S_\theta & r_z^2(1-\cos\theta) + C_\theta \end{bmatrix}$$

$\vec{r} \rightarrow \theta$

$$Ex \Rightarrow \vec{r} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = R_x$$

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} R_y \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow R_z$$

$$\begin{bmatrix} r_x r_z(1-\cos\theta) + r_y S_\theta \\ r_y r_z(1-\cos\theta) - r_x S_\theta \\ r_z^2(1-\cos\theta) + C_\theta \end{bmatrix}$$

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(3)

$$R = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \quad \underline{\text{Conhecido}}$$

$$\rightarrow \vec{r} \quad \theta?$$

$$r_{11} + r_{22} + r_{33} = r_x^2(1-\cos\theta) + \cos\theta +$$

$$r_y^2(1-\cos\theta) + \cos\theta +$$

$$r_z^2(1-\cos\theta) + \cos\theta$$

$$= \underbrace{(r_x^2 + r_y^2 + r_z^2)}_1 (1-\cos\theta) + 3\cos\theta$$

$$= 1 - \cos\theta + 3\cos\theta = 1 + 2\cos\theta$$

$$\checkmark \quad \cos\theta = \frac{r_{11} + r_{22} + r_{33} - 1}{2}$$

$$C^2 + S^2 = 1$$

$$\checkmark \quad S_\theta = \begin{pmatrix} + \\ - \end{pmatrix} \sqrt{1 - \cos^2\theta}$$

↓
escolher

$$\theta = \arctan 2(s_\theta, \cos\theta)$$

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④

$$r_{32} - r_{23} = 2 r_x s_\theta$$

$$r_{13} - r_{31} = 2 r_y s_\theta$$

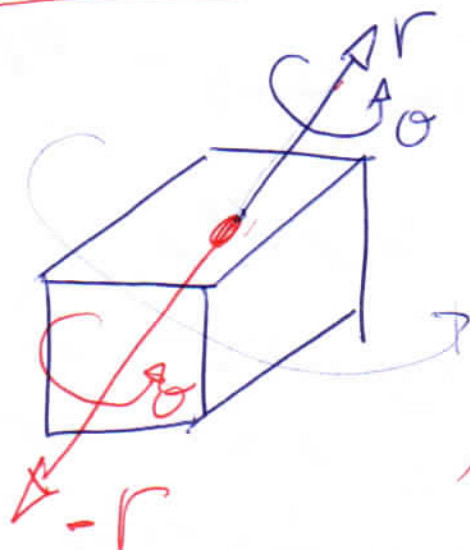
$$r_{21} - r_{12} = 2 r_z s_\theta$$

$$\vec{r} = \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} \frac{1}{2s_\theta}$$

$\Rightarrow$ base of
quaternions

Se $\boxed{s_\theta = 0} \Rightarrow \theta_r = 0 \text{ rad} \Rightarrow \omega = 1$
 $\theta_r = \pi \text{ rad} \Rightarrow \omega = -1$

Degeneration



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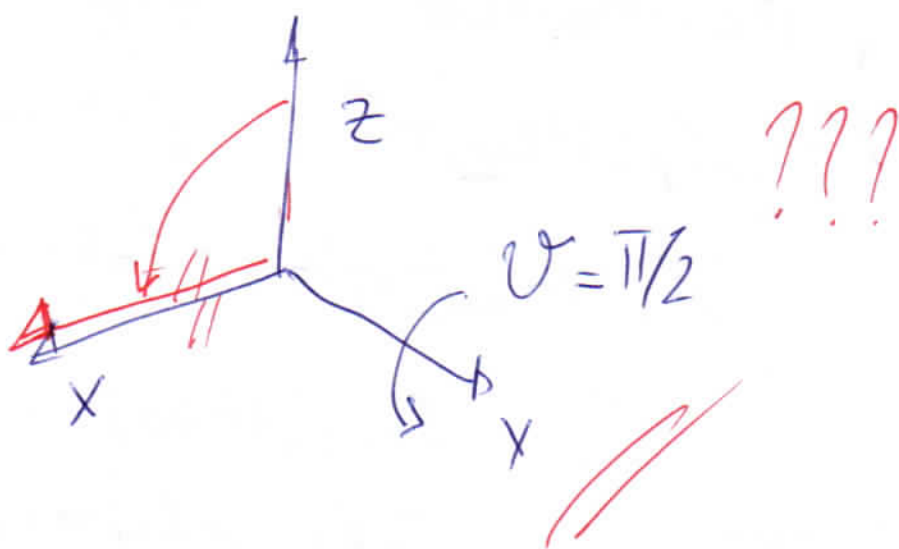
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$\Rightarrow$ No aula passada 30/11

2 RPY

Degeneração da matriz R_{RPY}

$$\text{Se } \boxed{C\psi = 0} \Rightarrow \begin{aligned} \sin\psi = +1 &= \psi = +\pi/2 \\ \sin\psi = -1 &= \psi = -\pi/2 \end{aligned}$$



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$$R = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \underline{\underline{E_{6\pi}}}$$

$$\Rightarrow \begin{aligned} \varphi &= \alpha \tan 2(\pm r_{23}, \pm r_{13}) \\ \vartheta &= \alpha \tan 2\left(\pm \sqrt{r_{13}^2 + r_{23}^2}, r_{33}\right) \\ \psi &= \alpha \tan 2\left(\pm r_{32}, \pm r_{31}\right) \end{aligned}$$

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1<sup>o</sup> solution  $sv > 0$

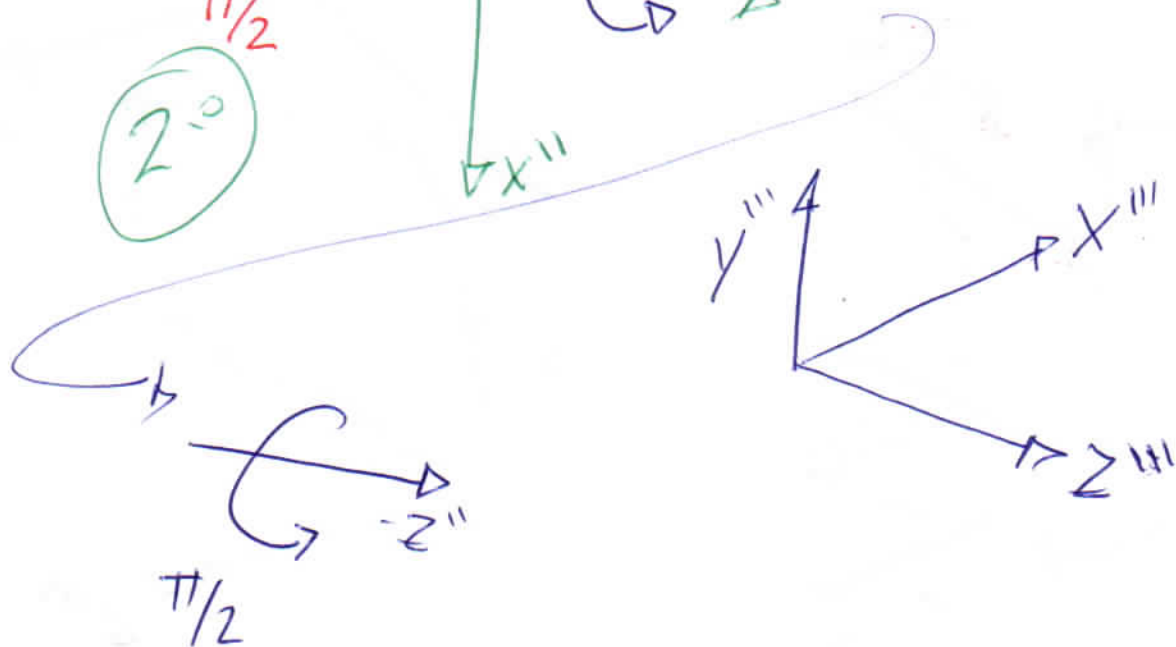
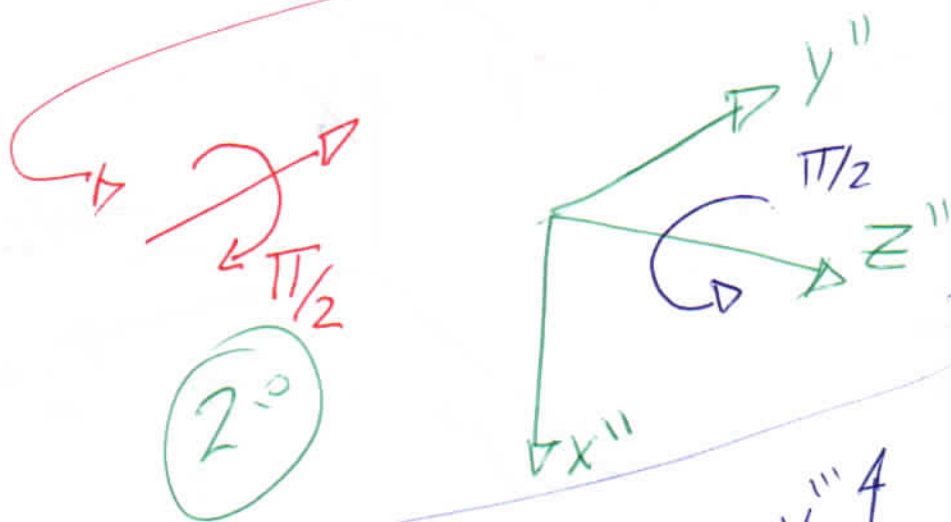
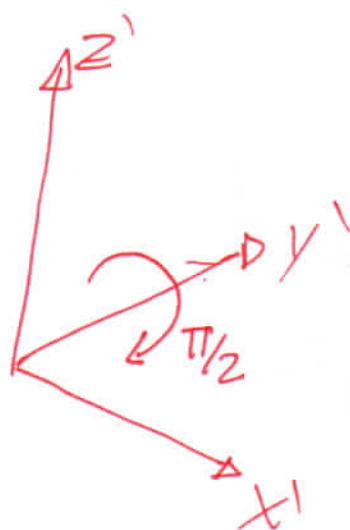
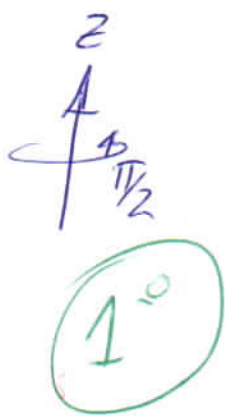
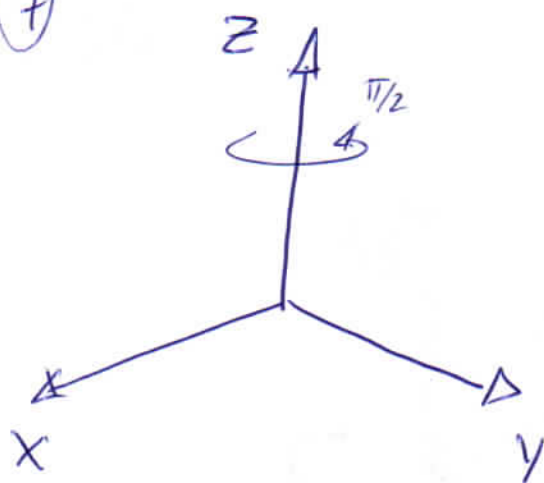
$$\begin{cases} \varphi = \alpha \tan 2(1, 0) = \pi/2 & z \\ \vartheta = \alpha \tan 2(+1, 0) = \pi/2 & y \checkmark \\ \psi = \alpha \tan 2(1, 0) = \pi/2 & z \end{cases}$$

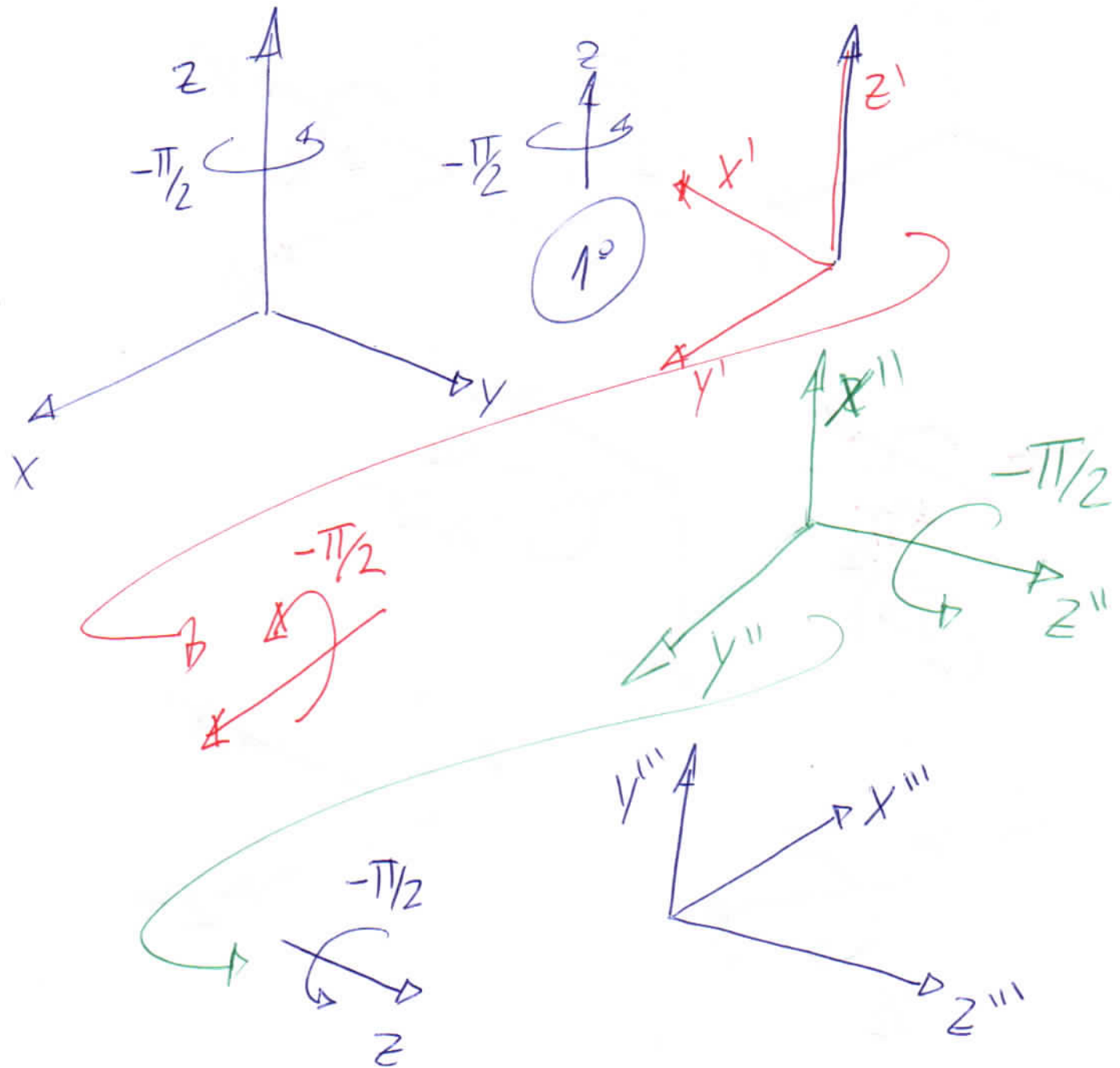
2<sup>o</sup> solution  $sv < 0$

$$\begin{cases} \varphi = \alpha \tan 2(-1, 0) = -\pi/2 & z \\ \vartheta = \alpha \tan 2(-1, 0) = -\pi/2 & y \\ \psi = \alpha \tan 2(-1, 0) = -\pi/2 & z \end{cases}$$

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Exercício =  $R \Rightarrow RPY$  2 soluções,  
 $\vec{r}, \vec{\theta}$